

केस स्टडी / CASE STUDY

OISD/CS/2022-23/POL/07

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INTRODUCTION

Title: Fire incident during loading of tank trucks at Tank lorry filling gantry resulting in fatalities.

Location: POL Depot

Result/ Outcome: 04 fatalities along with severe damage at bay no. 1, 2 & 3 of Tank lorry filling gantry.

BRIEF OF INCIDENT

- TLF and TWD operations were being done at the depot.
- Three tank trucks (TTs) were under filling at bay no. 1, 2 and 3 at Tank Lorry Filling Gantry (TLFG) of the depot.
- One TT was filling 4 KL MS & 8 KL HSD at bay no. 1 of TLFG. Suddenly a flash fire in the form of explosion occurred which engulfed TT at bay no. 1 and TT at bay no 2.
- At the time of fire, there were 08 persons at TLFG i.e., six TT crew and 02 contract workmen. Out of these 08 persons, 06 persons of TT crew and 01 contract workman were entrapped in the fire and got injured. One contract workman escaped unhurt.
- Emergency Shut Down (ESD) was activated from control room. Location in charge (LIC) had seen flash fire from his office and reached the site.
- Firefighting operation was started immediately by the location team. Communication was given to State fire department, mutual aid members, police department, local hospital regarding the emergency. The fire was extinguished within 03 minutes.

OBSERVATIONS & LAPSES

A. Safety measures at TLFG not followed/ ensured.

1. The quarterly safety checks for tank trucks of other OMCs were not being done.
2. The effective earthing of tank lorries during loading is not ensured. OMC TTs are issued the earthing boss/ ball as returnable item which is then fitted to TT by crew at TLFG without any supervision to ensure that a rigid and secured connection has been done. It was found fitted to the chassis of both other TTs.
3. The other earthing connection was found connected to painted steel guard of TT fuel tank.

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4. In case any excess quantity is to be withdrawn from the tank truck, unloading point was not being used as an approved arrangement. Instead of this, a metallic container was being used/ lowered in the filled compartment and the product was thus collected and stored in buckets. Any shortfall of the product was delivered with these buckets and funnel at the top of TT.
5. The tank lorry crew was not wearing the protective clothing i.e., cotton coverall which is a deemed basic PPE for all the personnel entering the operational area as per OISD-STD-155.
6. It was observed from the quarterly safety check report for TT involved in the incident that the electrical circuit is provided with double pole wiring. However, as per the preliminary investigation report of the company, TT was having single pole wiring.

B. Violation of the Petroleum Rules for transportation of petroleum products by tank lorry

1. Battery was not isolated from the electrical circuit by operating the master switch. The battery cut out switch of all 03 TTs was found in ON position.
2. As stated above, the earthing wire was not connected at designated points. The requirement of bonding at the top of TT is also not mentioned in the job details sheet provided to TLFG operators. It was found disconnected at the time of site visit.
3. The frequent lowering and raising of the dip rod by TT crew to check the product level during the loading operation was observed which result in generation of static charge and air vapour mixture at TLFG.
4. The fill pipe, emergency vent and dip pipe caps were not closed during loading of one compartment.

C. Nonadherence to SOPs

1. As per approved operations manual, location specific SOPs as applicable to TLF operations shall be followed in toto. However, many unsafe acts which result in an unsafe condition at TLGF were witnessed from the CCTV footage due to non-adherence to SOPs. None of the cameras of the CCTV system were properly covering the critical operating area i.e bay no. 1 and 8 at TLFG.
2. One camera installed on the back side of TLFG is neither identified nor configured and was not functional. CCTV monitoring station at the control room was not functional and no record for its breakdown/ maintenance was available. The record for monitoring of CCTV footages on daily basis as per the guidelines was not available.
3. The contract workmen (semi-skilled category) carry out the duties of TLF supervisor at TLFG without any authorization.
4. Further, the record for compliance w.r.t selection, deployment of proper skilled manpower for effective operation and maintenance at TLFG is not available. The deceased contract workman was working at depot for last 03 years for various assignments. However, for the present assignment, the record of I-card, induction training as per requirement of OISD-STD-244 is not available.

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5. Upon interaction with other contract workman present at TLFG during the fire incident, it was observed that there was lack of awareness on TLFG operations, use of dip rods to check loading correctness for all TTs, likely alarms and action taken for resolution during the loading operations.
6. Thus, as the workmen did not report the generation of process alarms to TLF in-charge for resolution as per the laid down guidelines for the same.
7. The lab result for the HSD sample drawn from TT involved in the incident confirmed that the sample had failed in density and flash point. The observed values indicate presence of MS in this compartment. Hence, the emptiness of TT was not ensured before filling which is required as per SOP. Any such mixing leads to generation of more flammable air vapour mixture.
8. The complete switch over from old to new server was done without any MOC approval. As per the work progress report of the location, the important work for graphic modification as per as-built P & ID was in progress. An approved plan to ensure the completion of all the activities as per the approved schedule was not available.
9. As the graphics were not available, the control room officer could not monitor the depot operation including, generation of any specific alarm on stoppage of BCU and loading at TLFG and its attendance from the Control Room.
10. Depot control room was not manned during the fire incident when TLF and TWD operations were being carried out. The control room officer had left the location at without any handing over and taking over to monitor the ongoing operations.

D. Deviation in functioning of terminal automation system

1. From the events and alarms log history, many instances of loading interruption due to the safety interlock of static earthing and alarms, "CONNECT EARTH" were generated. The observations for TT involved on the fire incident are also on similar lines.
2. As per the time stamp of stop command from ESD to BCUs and time of alarm for stoppage of batch at bay no. 1, batch had stopped before ESD command. There was time gap between TAS server and safety PLC because safety PLC was yet to be synchronized with TAS server as per the records.

Root cause for the incident

Loading operations with violation of SOP, unsafe practices and without ensuring a proper earthing connection resulted in generation of spark due to differential potential between the loading arm and fill pipe which caused ignition of flammable product air-vapour mixture at tank lorry filling gantry.

Recommendations

1. The quarterly checks of safety fittings for all tank trucks including TTs under hospitality arrangement shall be carried out and records maintained. Also, the daily inspection of all tank trucks shall be strengthened by way of random verification from CCTV footage by safety officer and record maintained.

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2. The exact sequence of operations must be followed for filling the tank trucks. Tank truck shall be checked to confirm that fire extinguisher is placed near TT, earthing and bonding has been connected and battery switch has been cut off by switching off the master switch by way of physical checks and records maintained for the same.
3. All locations shall provide capacitance-based earthing as safety interlock for loading/ unloading operations of tank trucks.
4. The handling of loose product at the TLF gantry during regular operations shall be eliminated. In case of any requirement, the product shall be handled only after due approval of location in charge by an approved system wherein product is not exposed in open atmosphere at any point of time.
5. The implementation of sound safety culture to promote safe practices including wearing of all applicable personnel protective equipment (PPEs), as per OISD-STD-155, amongst all sections of personnel at the location shall be done.
6. Organizations shall review their transportation contract and ensure implementation of PPEs including dress and its fabric in line with OISD-STD-155.
7. The hazards associated with tank trucks with single pole shall be identified and the suitable mitigation measures shall be implemented to ensure that all tank trucks are with double pole wiring.
8. Dip rod, if used, shall be lowered into the tank or compartment before loading of product starts and it shall not be completely raised above the liquid level during or within one minute of the completion of the tank truck loading.
9. To eliminate vapour generation, it is recommended that organization should work out strategy to convert all top loading tank lorry filling gantries to bottom loading with vapour recovery system in a time bound manner.
10. The organizations shall review HSE manual and SOPs in view of this investigation and necessary amendments and revisions shall be done.
11. The design of CCTV layout shall facilitate the coverage of all critical areas for continuous monitoring. CCTV monitoring stations shall be always functional for area surveillance and monitor the ongoing operations at the depot. Any unsafe condition shall be identified, and desired corrective action taken to avoid recurrence.
12. The system for identification of unsafe acts and conditions and its reporting by the operating personnel shall be strengthened at the organization level. A motivation scheme with appropriate recognition may be introduced for encouragement of the field personnel. The random verification of CCTV system shall be done to identify the unsafe acts and timely corrective action ensured.
13. The work allocation for all the workmen shall be reviewed to ensure that it is in line with the approved SOPs. Any other workmen shall not be deployed for the critical and important operation without proper authorization. A detailed review of the competency level of the workmen deployed shall be assessed for the work allotted to them.
14. All operating personnel including the contract workmen shall be given training including refresher course as per OISD-STD-154 and records maintained.
15. MOC approval for all the critical works shall be obtained. The approving authority shall ensure that any risk likely to occur due to this change shall be properly

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identified and efficient controls are in place through an appropriate approved system. The work shall be completed within the approved timelines.

16. The location in charge shall ensure that depot control room is manned on a continuous basis during operations and in emergency. The person shall only be allowed to leave once a reliever is positioned by the immediate officer and records for the same maintained.
17. A firm action plan shall be implemented to ensure time synchronization between the process and safety PLC. Also, CCTV system shall have the same timings to ensure identification of any event/ activity on real time basis.
18. A detailed review of the alarms (for meter overrun, earthing failure etc.) shall be carried out by control room officer/ planning officer to assess the reason. Loading shall commence only after an appropriate resolution.
19. Training to TT crew for disengagement from safety harness and escape from the top of TT shall be conducted and records maintained.
20. The required supervision by the concerned official for the work assigned to contract workmen shall be ensured.
21. The companies shall take cognizance of the above recommendations and ensure that the recommendations are implemented across the organization with time bound action plan.



JET FIRE EMANATING FROM TANK TRUK UNDER LOADING AT TLF BAY NO. 1

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DAMAGED TANK TRUCKS AT TLF GANTRY



DAMAGE FOR ELECTRICAL & INSTRUMENTATION SYSYTEM AT TLF GANTRY

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DAMAGED TT INVOLVED IN THE INCIDENT

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